

Will Piwowarski

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EDUCATION

Liberty University

B.S., Computer Science - Data Science & Software Engineering cognates (GPA: 3.91/4.0)

Aug 2024 - May 2028

Lynchburg, VA

- **Coursework:** Data Structures & Algorithms (C++), Programming in C++ I & II, Computer Architecture, Applied Linear Algebra, Probability & Statistics, Statistical Analysis, Discrete Mathematics

EXPERIENCE

RhobotAI | Software Engineering Intern

May 2026 - Present

- Build a three-stage streaming time-series pipeline that filters, aligns, and curates live CSV, MQTT, and SQL sensor streams into clean per-batch datasets in InfluxDB using MAD-based robust outlier detection and rolling steady-state filtering.
- Engineer a numerical optimization service that trains 20+ regression models, including CatBoost, XGBoost, LightGBM, and Random Forest, in parallel subprocesses before applying SciPy L-BFGS-B to fitted surrogates for yield-maximizing parameters scored with R² and RMSE.
- Audit the legacy prototype for correctness and security defects, driving its rebuild into a containerized, authenticated platform through object-oriented programming practices.
- Migrate the Flask ingestion service to FastAPI in a Dockerized stack with Vue 3, PostgreSQL, InfluxDB, MongoDB, and Redis, securing API development with JWT SSO.

i7 Consulting Group | Software Engineering Intern

May 2025 - Aug 2025

- Replaced an error-prone Excel workflow with a Vue 3 Composition API and Vuetify SPA built around an editable month-grid, along with CRUD administration screens.
- Designed a normalized nine-table MySQL schema with an append-only audit model tracking a save, submit, and approve lifecycle; exposed the workflow through a Spring Boot Java 17 REST API using Spring Data JPA and documented it with Swagger/OpenAPI.
- Built an Apache POI engine generating approval-gated Excel invoices with live formulas, plus Spring Security authentication with BCrypt password hashing.
- Packaged the SPA as a multi-stage Docker image served by nginx, reverse-proxying the API and enforcing mutual-TLS client-certificate authentication.

PROJECTS

Flock | getflock.cc

- Architected a multi-tenant PWA with per-tenant data isolation in depth: a server-side scoped data layer injects the tenant ID into every query, a role × resource × scope permission model rides in JWT claims, and Postgres row-level security backs both.
- Built a hybrid attendance system with self-check-in, leader confirmation, and hall-wide rollups, adding rule-based trend analytics and at-risk flagging to an offline-capable installable PWA deployed to 200+ students across 3 residence halls.

Cartograph | github.com/willpiwowarski

- Built a TypeScript static-analysis CLI that constructs a directed call graph by resolving each call through the type checker (ts-morph), disambiguating same-named methods by receiver type, aliased imports, and inheritance overrides that text search misses.
- Modeled the graph as bidirectional adjacency maps for constant-time caller/callee lookups and BFS impact analysis: direct callers, transitive reachability, and shortest call path between two functions.
- Enforced correctness with a hand-traced regression suite run in GitHub Actions CI on pushes and PRs to main, failing the build on any call-graph divergence.

EXTRACURRICULAR

AI Club - SkyWatch AI Challenge (with Xenith Solutions) | Technical Lead

Aug 2026 - Present

- Leading a cross-functional student team building a real-time aircraft-surveillance pipeline: ingesting live ADS-B broadcasts via SDR/Raspberry Pi into a cloud Kafka pipeline, applying ML to model airspace pattern-of-life and flag anomalies.
- Own hardware, live data feed, integration, and repository discipline across hardware, data, and ML sub-teams; coordinate weekly with the industry sponsor and mentors from Google and Framatome.

Liberty University Residence Life | Resident Shepherd

Aug 2025 - Present

- Oversee a residence hall of ~70 students and 9–10 community group leaders, coordinating weekly meetings, one-on-one mentorship, and per-semester written reporting.

SKILLS

- **Languages:** C++, Python, Java, TypeScript, SQL (Postgres, MySQL), JavaScript, HTML/CSS, Object-Oriented Programming
- **Data & ML:** NumPy, pandas, SciPy (L-BFGS-B optimization), scikit-learn, CatBoost, XGBoost, LightGBM, InfluxDB
- **Frameworks & Libraries:** FastAPI, Pydantic, Spring Boot, React, Next.js, Node.js, Vue 3, Prisma, Tailwind CSS, API Development
- **Cloud & DevOps:** Docker, AWS (S3, IAM), Redis, MongoDB, Git, GitHub Actions, Vercel, Neon
- **AI:** Google Gemini, LLM output validation, prompt engineering